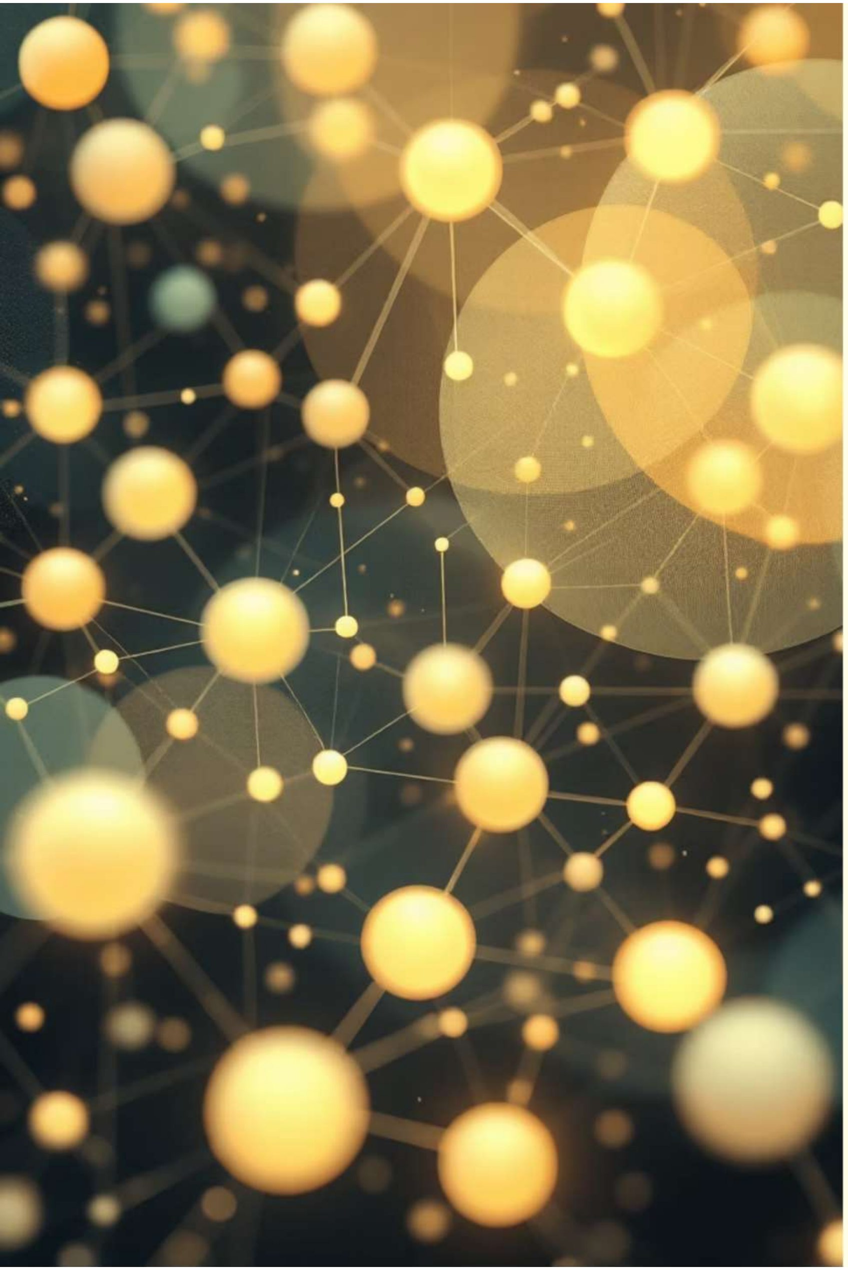


SPS Research Framework

Supervisory Prioritisation Score for Retail Payment Systems

Pietro Maietta · University of Naples "Parthenope" · Innovation Data Challenge 2026 – NayaOne Sandbox

MARCH 2026 | DORA EU 2022/2554 | PSD2 EU 2015/2366

Notebook Structure

Eight sequential phases transform raw transaction data into actionable supervisory intelligence.

01

Data Ingestion & Topological Analysis

GLM instrument vulnerability deep dive

02

AutoML Genetic Optimiser

Calibrate α (technical) and β (social) weights

03

SPS Recalculation & Multi-Algorithm ML

XGBoost, Random Forest, Logistic Regression

04

Dashboard & Vigilance Matrix

4-class PSP segmentation and validation

05

Network Topology + Final Validation

DORA stress test & PSD2 heatmap

06

Time-Series Surveillance

ARIMA (744 h) + SARIMA (12 months)

Phase 1 – Data Ingestion & Fusion

Dataset

414K	698	20
Transactions	PSPs	Columns

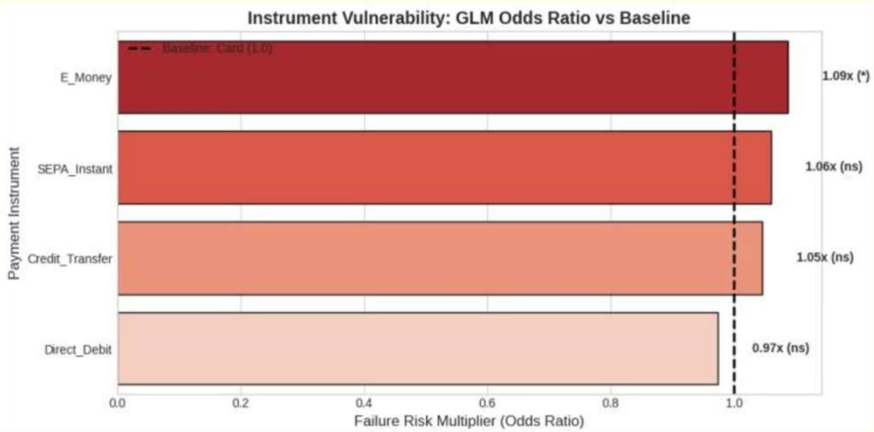
Data Sources (NayaOne API)

- D01 Transactions · D06 PSP Registry
- D03 Performance · D07 Channels · D10 Profiles

Missing Data Handling

34.19% of `processing_time_ms` missing (Completed transactions only). Imputed via per-instrument median — Card: 1,479 ms · Credit Transfer: 5,412,169 ms · Direct Debit: 5,420,446 ms.

Phase 1.5 – GLM Instrument Vulnerability



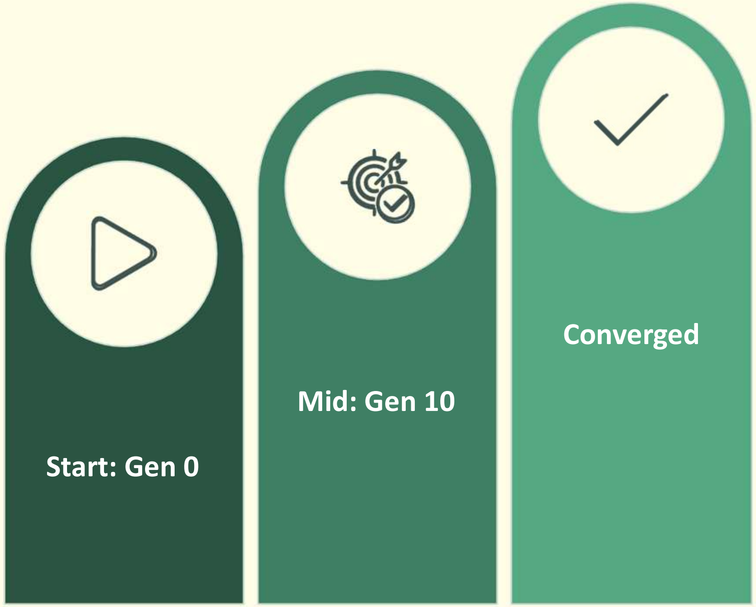
Logistic Regression vs Card Baseline

E_Money is the only instrument with a statistically significant higher failure risk (OR = 1.087, p = 0.040). SEPA_Instant and Credit_Transfer show marginal elevation but are non-significant. Direct_Debit is marginally below baseline.

Real Failure Rates

Global mean: **2.02%**. E_Money leads at 2.16%, Direct_Debit lowest at 1.94% — a narrow band confirming overall system stability.

Phase 2 – AutoML Genetic Optimiser



Convergence reached at generation 10 – fitness stable for 10 consecutive generations (std = 0.000000).

Optimisation Results

α (Technical Weight)

0.2404 – DORA operational resilience

β (Social Weight)

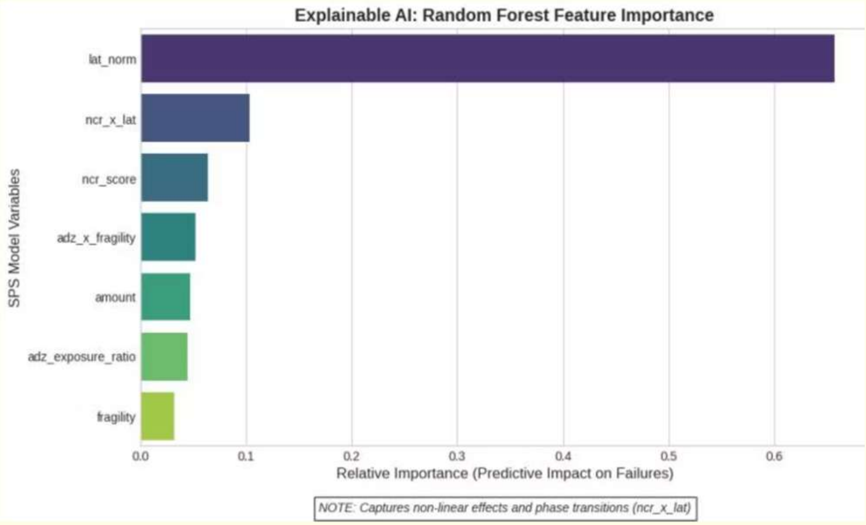
0.2404 – PSD2 financial inclusion

Ratio α/β

1.00:1 – Balanced bi-focal risk

Vigilance Diagnosis: Equal weights confirm prevalence of **technical risks** (hub failure). Priority: DORA inspections on high-NCR hubs.

Phase 3 – Machine Learning Inference



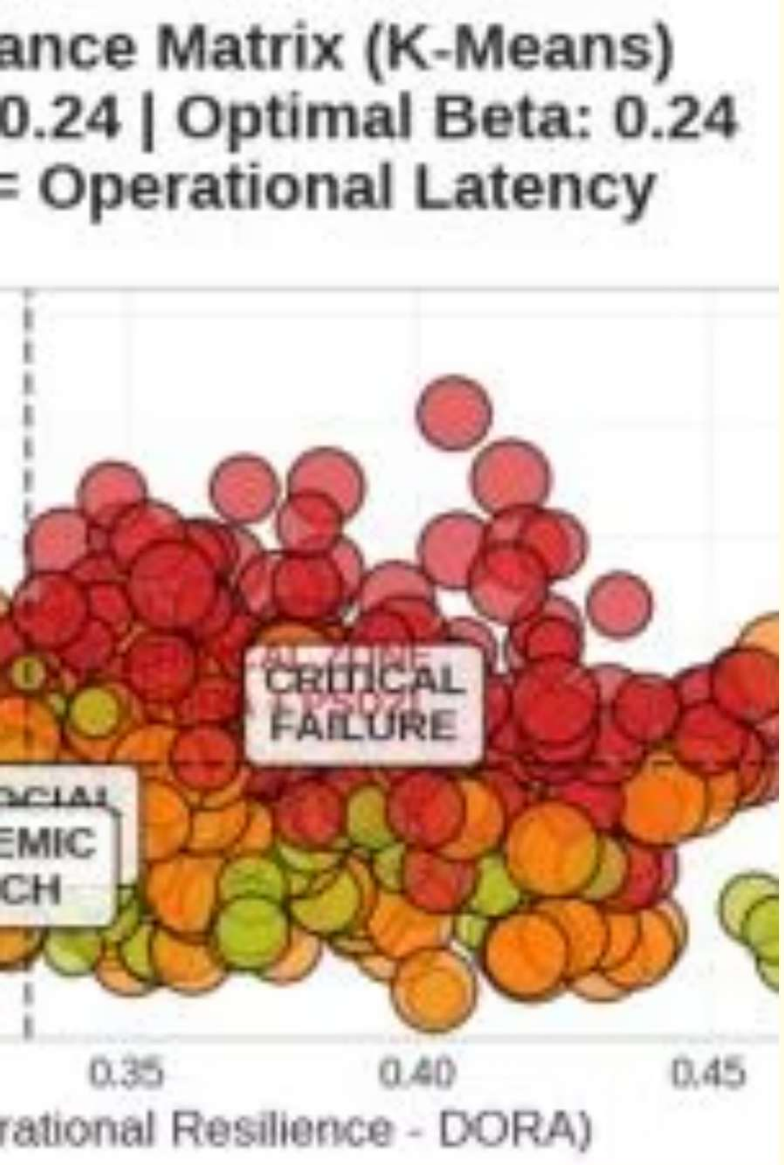
Multi-Algorithm Comparison

Algorithm	ROC-AUC	F1
XGBoost	0.6855 ✓	0.0618
Random Forest	0.6845	0.0599
Logistic Regression	0.5038	0.0000

Explainable AI – Top Features

- lat_norm: 65.7% – operational congestion is the main driver
- ncr_x_lat: 10.4% – confirms NCR × Latency phase transitions
- ncr_score: 6.4% – network centrality matters

Non-linearities captured by XGBoost validate the evolutionary SPS approach. PR-AUC is the most informative metric for this imbalanced use-case (2.02% minority class).



Phase 4 – Vigilance Matrix

Critical Failure

201 PSPs (28.8%)SPS mean: 0.1393 · NCR mean: 0.6356
→ Immediate DORA inspection

Social Risk

167 PSPs (23.9%)SPS mean: 0.1206 · Social: 0.1666
→ PSD2 inclusion audit

Systemic Watch

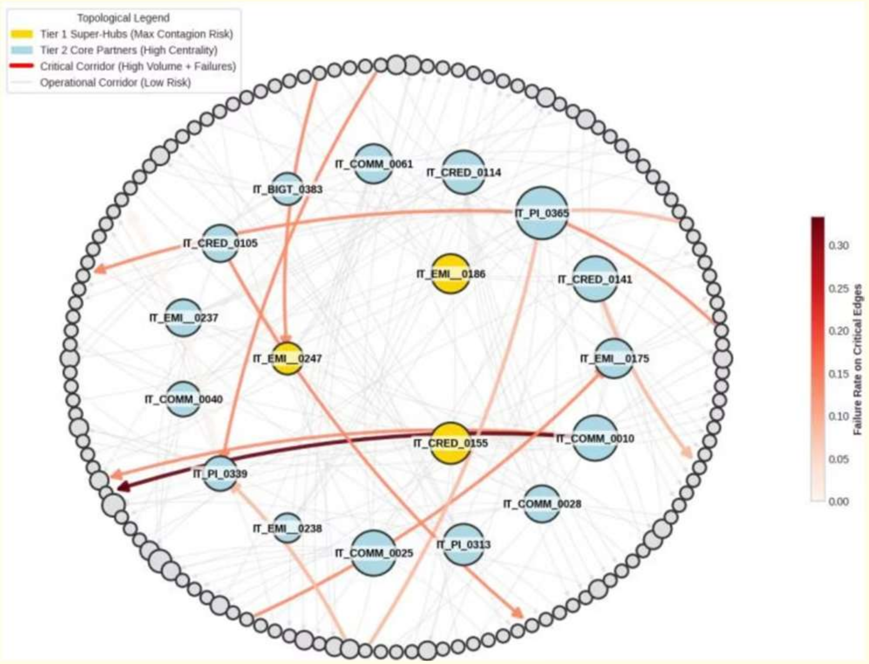
159 PSPs (22.8%)SPS mean: 0.1166
→ Continuous monitoring

Standard

171 PSPs (24.5%)SPS mean: 0.1073 · NCR mean: 0.2792
→ No action required

K-Means clustering is validated: SPS ranking follows the expected hierarchy CRITICAL > SOCIAL > SYSTEMIC > STANDARD with a 0.0320-point spread between extremes.

Phase 5 – Network Topology



Scale-Free Topology

Power-law exponent $\Gamma = 2.7892$, aligned with the Barabási-Albert model – confirming the network is scale-free and vulnerable to targeted hub removal.

Tier 1 Super-Hubs (Max Betweenness) Betweenness)

DORA Stress Test

All three scenarios (single, triple, systemic shock) returned **Fragmentation Index = 0.0** and **Blackout = 0%** – network shows good topological resilience. Recovery Complexity: 5.

IT_EMI__0186

NCR = 1.0000

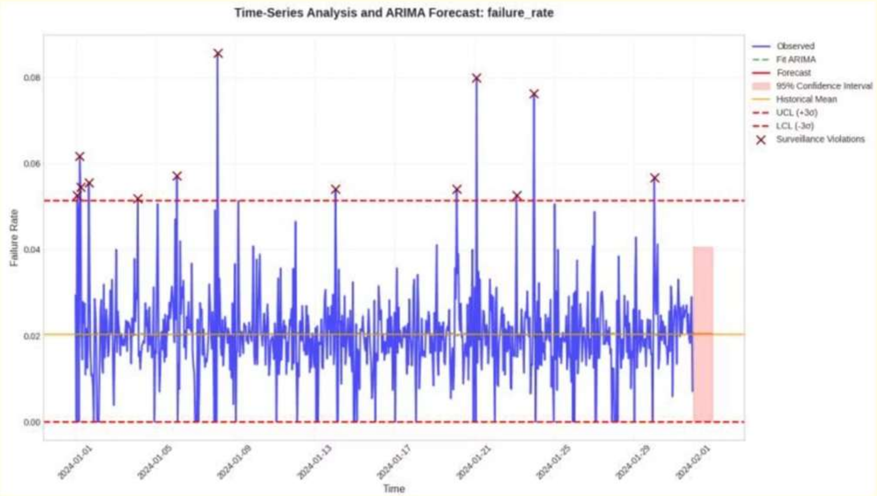
IT_EMI__0247

NCR = 0.9002

IT_CRED_0155

NCR = 0.8867

Phase 7 & 8 – Time-Series Surveillance



ARIMA – Hourly (Phase 7)

- **744 hourly periods** analysed (Jan 2024)
- All three series stationary (ADF p = 0.000)
- ARIMA(2,0,2) fitted for failure_rate, avg_latency, avg_ncr
- Surveillance violations flagged where data exceeds UCL (3σ)

SARIMA – Monthly (Phase 8)

- **12 months** of data, ~400K–430K transactions/month
- Coefficient of Variation <10% for SPS and NCR → STABLE
- SARIMA confirms absence of critical seasonal trends

Key Results & Supervisory Recommendations

DORA Priority

201 PSPs (28.8%) require immediate inspection — focus on latency and hub congestion

PSD2 Priority

167 PSPs (23.9%) require inclusion audit — focus on Active Denial Zones (ADZ)

Scale-Free Network

Γ = 2.7892 · 3 Tier-1 super-hubs identified · resilience confirmed under stress

Predictive ML

XGBoost ROC-AUC = 0.6855 · lat_norm dominates (65.7%) · non-linearities validated

Bi-focality validated: Both DORA (operational resilience) and PSD2 (financial inclusion) risks are material. The SPS framework is ready for submission to the Innovation Data Challenge 2026 and presentation to supervisory authorities.